$\mu = np$ where n represents the number of independent trials and p represents the probability of success in a single trial.

Practice

4.1 Probability Distribution Function (PDF) for a Discrete Random Variable

Use the following information to answer the next five exercises: A company wants to evaluate its attrition rate, in other words, how long new hires stay with the company. Over the years, they have established the following probability distribution.

Let X = the number of years a new hire will stay with the company.

Let $P(x)$ = the probability that a new hire will stay with the company x years.

1. Complete Table 4.20 using the data provided.

x	$P(x)$
0	0.12
1	0.18
2	0.30
3	0.15
4	
5	0.10
6	0.05

Table 4.20

2. $P(x = 4)$ = _____
3. $P(x \geq 5)$ = _____
4. On average, how long would you expect a new hire to stay with the company?
5. What does the column "$P(x)$" sum to?

Use the following information to answer the next six exercises: A baker is deciding how many batches of muffins to make to sell in his bakery. He wants to make enough to sell every one and no fewer. Through observation, the baker has established a probability distribution.

x	$P(x)$
1	0.15
2	0.35
3	0.40
4	0.10

Table 4.21

6. Define the random variable X.

7. What is the probability the baker will sell more than one batch? $P(x > 1) =$ _____
8. What is the probability the baker will sell exactly one batch? $P(x = 1) =$ _____
9. On average, how many batches should the baker make?

Use the following information to answer the next four exercises: Ellen has music practice three days a week. She practices for all of the three days 85% of the time, two days 8% of the time, one day 4% of the time, and no days 3% of the time. One week is selected at random.

10. Define the random variable X.
11. Construct a probability distribution table for the data.
12. We know that for a probability distribution function to be discrete, it must have two characteristics. One is that the sum of the probabilities is one. What is the other characteristic?

Use the following information to answer the next five exercises: Javier volunteers in community events each month. He does not do more than five events in a month. He attends exactly five events 35% of the time, four events 25% of the time, three events 20% of the time, two events 10% of the time, one event 5% of the time, and no events 5% of the time.

13. Define the random variable X.
14. What values does x take on?
15. Construct a PDF table.
16. Find the probability that Javier volunteers for less than three events each month. $P(x < 3) =$ _____
17. Find the probability that Javier volunteers for at least one event each month. $P(x > 0) =$ _____

4.2 Mean or Expected Value and Standard Deviation

18. Complete the expected value table.

x	$P(x)$	$x*P(x)$
0	0.2	
1	0.2	
2	0.4	
3	0.2	

Table 4.22

19. Find the expected value from the expected value table.

x	$P(x)$	$x*P(x)$
2	0.1	2(0.1) = 0.2
4	0.3	4(0.3) = 1.2
6	0.4	6(0.4) = 2.4
8	0.2	8(0.2) = 1.6

Table 4.23

20. Find the standard deviation.

x	$P(x)$	$x*P(x)$	$(x-\mu)^2P(x)$
2	0.1	2(0.1) = 0.2	$(2–5.4)^2(0.1) = 1.156$
4	0.3	4(0.3) = 1.2	$(4–5.4)^2(0.3) = 0.588$
6	0.4	6(0.4) = 2.4	$(6–5.4)^2(0.4) = 0.144$
8	0.2	8(0.2) = 1.6	$(8–5.4)^2(0.2) = 1.352$

Table 4.24

21. Identify the mistake in the probability distribution table.

x	$P(x)$	$x*P(x)$
1	0.15	0.15
2	0.25	0.50
3	0.30	0.90
4	0.20	0.80
5	0.15	0.75

Table 4.25

22. Identify the mistake in the probability distribution table.

x	$P(x)$	$x*P(x)$
1	0.15	0.15
2	0.25	0.40
3	0.25	0.65
4	0.20	0.85
5	0.15	1

Table 4.26

Use the following information to answer the next five exercises: A physics professor wants to know what percent of physics majors will spend the next several years doing post-graduate research. They have the following probability distribution.

x	$P(x)$	$x*P(x)$
1	0.35	
2	0.20	
3	0.15	
4		
5	0.10	
6	0.05	

Table 4.27

23. Define the random variable X.

24. Define $P(x)$, or the probability of x.

25. Find the probability that a physics major will do post-graduate research for four years. $P(x = 4)$ = _____

26. FInd the probability that a physics major will do post-graduate research for at most three years. $P(x \leq 3)$ = _____

27. On average, how many years would you expect a physics major to spend doing post-graduate research?

Use the following information to answer the next seven exercises: A ballet instructor is interested in knowing what percent of each year's class will continue on to the next, so that she can plan what classes to offer. Over the years, she has established the following probability distribution.

- Let X = the number of years a student will study ballet with the teacher.
- Let $P(x)$ = the probability that a student will study ballet x years.

28. Complete Table 4.28 using the data provided.

x	$P(x)$	$x*P(x)$
1	0.10	
2	0.05	
3	0.10	
4		
5	0.30	
6	0.20	
7	0.10	

Table 4.28

29. In words, define the random variable X.

30. $P(x = 4)$ = _____

31. $P(x < 4)$ = _____

32. On average, how many years would you expect a child to study ballet with this teacher?

33. What does the column "$P(x)$" sum to and why?

34. What does the column "$x*P(x)$" sum to and why?

35. You are playing a game by drawing a card from a standard deck and replacing it. If the card is a face card, you win $30. If it is not a face card, you pay $2. There are 12 face cards in a deck of 52 cards. What is the expected value of playing the game?

36. You are playing a game by drawing a card from a standard deck and replacing it. If the card is a face card, you win $30. If it is not a face card, you pay $2. There are 12 face cards in a deck of 52 cards. Should you play the game?

4.3 Binomial Distribution

Use the following information to answer the next eight exercises: The Higher Education Research Institute at UCLA collected data from 203,967 incoming first-time, full-time first-year students from 270 four-year colleges and universities in the U.S. 71.3% of those students replied that, yes, they believe that same-sex couples should have the right to legal marital status. Suppose that you randomly pick eight first-time, full-time first-year students from the survey. You are interested in the number that believes that same sex-couples should have the right to legal marital status.

37. In words, define the random variable X.

38. X ~ _____(_____,_____)

39. What values does the random variable X take on?

40. Construct the probability distribution function (PDF).

x	$P(x)$

Table 4.29

41. On average (μ), how many would you expect to answer yes?

42. What is the standard deviation (σ)?

43. What is the probability that at most five of the first-year students reply “yes”?

44. What is the probability that at least two of the first-year students reply “yes”?

4.4 Geometric Distribution

Use the following information to answer the next six exercises: The Higher Education Research Institute at UCLA collected data from 203,967 incoming first-time, full-time first-year students from 270 four-year colleges and universities in the U.S. 71.3% of those students replied that, yes, they believe that same-sex couples should have the right to legal marital status. Suppose that you randomly select first-year students from the study until you find one who replies “yes.” You are interested in the number of first-year students you must ask.

45. In words, define the random variable X.

46. X ~ _____(_____,_____)

47. What values does the random variable X take on?

48. Construct the probability distribution function (PDF). Stop at $x = 6$.

x	$P(x)$
1	
2	
3	
4	
5	
6	

Table 4.30

49. On average (μ), how many first-year students would you expect to have to ask until you found one who replies "yes?"

50. What is the probability that you will need to ask fewer than three first-year students?

4.5 Hypergeometric Distribution

Use the following information to answer the next five exercises: Suppose that a group of statistics students is divided into two groups: business majors and non-business majors. There are 16 business majors in the group and seven non-business majors in the group. A random sample of nine students is taken. We are interested in the number of business majors in the sample.

51. In words, define the random variable X.

52. $X \sim$ _____(_____,_____)

53. What values does X take on?

54. Find the standard deviation.

55. On average (μ), how many would you expect to be business majors?

4.6 Poisson Distribution

Use the following information to answer the next six exercises: On average, a clothing store gets 120 customers per day.

56. Assume the event occurs independently in any given day. Define the random variable X.

57. What values does X take on?

58. What is the probability of getting 150 customers in one day?

59. What is the probability of getting 35 customers in the first four hours? Assume the store is open 12 hours each day.

60. What is the probability that the store will have more than 12 customers in the first hour?

61. What is the probability that the store will have fewer than 12 customers in the first two hours?

62. Which type of distribution can the Poisson model be used to approximate? When would you do this?

Use the following information to answer the next six exercises: On average, eight teens in the U.S. die from motor vehicle injuries per day. As a result, states across the country are debating raising the driving age.

63. Assume the event occurs independently in any given day. In words, define the random variable X.

64. $X \sim$ _____(_____,_____)

65. What values does X take on?

66. For the given values of the random variable X, fill in the corresponding probabilities.

67. Is it likely that there will be no teens killed from motor vehicle injuries on any given day in the U.S? Justify your answer numerically.

68. Is it likely that there will be more than 20 teens killed from motor vehicle injuries on any given day in the U.S.? Justify your answer numerically.

Homework

4.1 Probability Distribution Function (PDF) for a Discrete Random Variable

69. Suppose that the PDF for the number of years it takes to earn a Bachelor of Science (B.S.) degree is given in Table 4.31.

x	$P(x)$
3	0.05
4	0.40
5	0.30
6	0.15
7	0.10

Table 4.31

a. In words, define the random variable X.
b. What does it mean that the values zero, one, and two are not included for x in the PDF?

4.2 Mean or Expected Value and Standard Deviation

70. A theater group holds a fund-raiser. It sells 100 raffle tickets for $5 apiece. Suppose you purchase four tickets. The prize is two passes to a Broadway show, worth a total of $150.
a. What are you interested in here?
b. In words, define the random variable X.
c. List the values that X may take on.
d. Construct a PDF.
e. If this fund-raiser is repeated often and you always purchase four tickets, what would be your expected average winnings per raffle?

71. A game involves selecting a card from a regular 52-card deck and tossing a coin. The coin is a fair coin and is equally likely to land on heads or tails.
- If the card is a face card, and the coin lands on Heads, you win $6
- If the card is a face card, and the coin lands on Tails, you win $2
- If the card is not a face card, you lose $2, no matter what the coin shows.

a. Find the expected value for this game (expected net gain or loss).
b. Explain what your calculations indicate about your long-term average profits and losses on this game.
c. Should you play this game to win money?

72. You buy a lottery ticket to a lottery that costs $10 per ticket. There are only 100 tickets available to be sold in this lottery. In this lottery there are one $500 prize, two $100 prizes, and four $25 prizes. Find your expected gain or loss.

73. Complete the PDF and answer the questions.

x	$P(x)$	$xP(x)$
0	0.3	
1	0.2	
2		
3	0.4	

Table 4.32

a. Find the probability that $x = 2$.
b. Find the expected value.

74. Suppose that you are offered the following "deal." You roll a die. If you roll a six, you win $10. If you roll a four or five, you win $5. If you roll a one, two, or three, you pay $6.
a. What are you ultimately interested in here (the value of the roll or the money you win)?
b. In words, define the Random Variable X.
c. List the values that X may take on.
d. Construct a PDF.
e. Over the long run of playing this game, what are your expected average winnings per game?
f. Based on numerical values, should you take the deal? Explain your decision in complete sentences.

75. A venture capitalist, willing to invest $1,000,000, has three investments to choose from. The first investment, a software company, has a 10% chance of returning $5,000,000 profit, a 30% chance of returning $1,000,000 profit, and a 60% chance of losing the million dollars. The second company, a hardware company, has a 20% chance of returning $3,000,000 profit, a 40% chance of returning $1,000,000 profit, and a 40% chance of losing the million dollars. The third company, a biotech firm, has a 10% chance of returning $6,000,000 profit, a 70% of no profit or loss, and a 20% chance of losing the million dollars.
a. Construct a PDF for each investment.
b. Find the expected value for each investment.
c. Which is the safest investment? Why do you think so?
d. Which is the riskiest investment? Why do you think so?
e. Which investment has the highest expected return, on average?

76. Suppose that 20,000 married adults in the United States were randomly surveyed as to the number of children they have. The results are compiled and are used as theoretical probabilities. Let X = the number of children married people have.

x	$P(x)$	$xP(x)$
0	0.10	
1	0.20	
2	0.30	
3		
4	0.10	

Table 4.33